

RibAssist 3D: Biplanar Rib-Fracture Detection, Addressing, and Selective 3D Localization from CT-Derived Projections

An assistive-workflow proof of concept, confirmed on a sealed test cohort

Kabila Haile Soboka 
The University of Texas at Austin
Austin, Texas, USA
kabilahailesoboka@gmail.com

ABSTRACT

Rib fractures are common, clinically significant, and time-consuming to localize on computed tomography (CT). This project asks a narrow, mechanistic question: can fractures detected independently in two orthogonal projections (anteroposterior, AP, and lateral) be paired across views and triangulated into reliable 3D fracture points at a controlled rate of false 3D outputs? We answer it with a staged diagnostic study rather than a single end-to-end number. First, the biplanar geometry is exact and, given correct correspondence, highly accurate: back-projection of correct paired centers gives 0.0 mm round-trip error, and detector-predicted centers reconstruct to median 4.0 mm 3D error with 88% within 10 mm and 93.6% rib-exact. Second, on the sealed cohort a substantial share of fractures is in principle recoverable: dual-view availability reaches 61.1% and the candidate graph contains a correct pair for 58.4% of fractures. The binding limitation is therefore neither geometry nor localization but confidence-limited cross-view correspondence: on the frozen detector, both deterministic and learned pair-scoring reconstruct 0% of fractures at a budget of one false 3D point per case. A controlled factorial attributes the operational gain to lateral-detector quality rather than to the tested deterministic or local-appearance matchers; retraining the lateral head lifts dual-view availability (0.52 to 0.76 in development) and moves the frontier from 0% to 2.44% recall at 10 mm, driven by deterministic detector-confidence assignment. When the policy commits a correct pair, the emitted point is geometrically accurate (sealed median 1.49 mm, rib-exact 93%). As the terminal operating point of a deliberately conservative fixed policy, a single pre-specified pass on the untouched 55-case cohort promotes 15 of 601 fractures to correct 3D localizations at 0.436 false 3D points per case: a 2.50% end-to-end commitment yield (case-bootstrap 95% interval [0.69%, 4.48%], excludes zero), versus 0% for the original detector’s selected controlled-budget policy. This end-to-end yield is the intersection of several selective gates, not a measure of the detector, geometry, or rib addressing in isolation. The contribution is best read as validated biplanar reconstruction geometry with high conditional localization fidelity, a staged identification of cross-view correspondence confidence as the effective bottleneck, and a selective assistive workflow that preserves uncertain findings, rather than as a standalone automatic reconstructor.

KEYWORDS

rib fracture, computed tomography, biplanar reconstruction, cross-view correspondence, selective localization, assistive re-view workflow, abstention, medical imaging, proof of concept

1 INTRODUCTION

Rib fractures are among the most frequently encountered injuries in thoracic trauma, and increasing fracture burden is associated with greater morbidity and mortality; fracture count and the presence of flail segments therefore contribute to clinical risk assessment and management decisions [1, 8, 11]. Yet localizing every fracture on a CT volume is tedious and error-prone; subtle non-displaced fractures are easily missed on axial slices, and manually assigning each finding to an anatomical rib level (“addressing”) is slow. Automated 3D localization would relieve this burden, but full volumetric detection networks are computationally heavy and require dense 3D annotation.

An attractive alternative is a projection-based route: render the CT to two orthogonal orthographic views, detect fractures independently in each, and triangulate the matched detections back into 3D. A projection-based approach offers a lower-dimensional alternative that may reduce inference and annotation complexity relative to dense volumetric detection. The route hinges on one hard sub-problem: solving the cross-view correspondence, i.e., deciding which AP detection and which lateral detection are the same physical fracture, at a controlled rate of false 3D outputs. Because the deployed object is a 3D point, success must be measured operationally, never using hidden ground-truth correspondence identity.

Crucially, the AP and lateral views used here are deterministic orthographic renderings of CT volumes, not independently acquired clinical radiographs. The study therefore evaluates the reconstruction mechanism under controlled geometry rather than generalization to real biplanar radiography, and any clinical framing below should be read with that scope in mind.

This paper reports a staged diagnostic study of exactly that sub-problem, built on these simulated biplanar projections derived from the public RibFrac and RibSeg datasets. Rather than claiming a deployable detector, we treat the system as an object of investigation and ask precisely where a biplanar 3D rib-fracture reconstructor succeeds and where it fails. Our contributions are: (1) a decomposition of the pipeline into geometry, localization, and correspondence, with a controlled

experiment isolating each; (2) evidence that the operational gain is attributable to lateral-detector quality rather than to the tested deterministic or local-appearance correspondence methods; (3) a lateral retraining intervention that produces the first nonzero controlled-budget reconstructions; and (4) a single pre-specified confirmation on an untouched sealed cohort, with fail-closed provenance, that reproduces the development estimate out of sample.

RibAssist 3D is evaluated primarily as a confidence-gated 3D reconstruction system, but its utility is broader than automatic triangulation: even when correspondence cannot be committed, the per-view detections still highlight suspicious regions and support rib-level addressing. The workflow therefore preserves the 2D findings and treats 3D localization as an additive output, so the system’s value does not collapse to zero when it abstains. We distinguish two readings throughout: a *standalone automatic reconstructor*, which the evidence does not yet support, and an *assistive highlighting, addressing, and selective-localization system*, which it supports as a proof of concept.

2 RELATED WORK

Deep rib-fracture detection on CT. The closest reference point is FracNet [5], which established the RibFrac benchmark and a 3D U-Net-style detector that segments and detects rib fractures directly in the CT volume. FracNet operates fully in 3D and reports free-response operating characteristic (FROC) sensitivity against radiologist labels. Our work shares the detection target and the FROC-style, budget-aware evaluation philosophy, but deliberately departs from volumetric inference: we detect in 2D projections and study whether the cheaper biplanar route can recover 3D points at a controlled false rate. FracNet also motivates our operational endpoint, since a clinical 3D CAD system is judged by how many true fractures it localizes per allowed false finding, not by pixel overlap.

Rib labeling and anatomical centerlines. Assigning a fracture to a specific rib requires rib identity. RibSeg v2 [4] provides large-scale rib labels and anatomical centerlines and benchmarks point-cloud methods for rib segmentation and centerline extraction. We use RibSeg-derived rib geometry both as the substrate for our “addressing” model, which predicts side and rib level for each detection, and as the anatomical ground truth against which reconstructed points are checked for rib-exactness. Where RibSeg v2 treats rib parsing as the end goal, we consume it as a supporting signal for fracture localization and as an evaluation oracle.

Biplanar-to-3D reconstruction. The idea of recovering 3D structure from two orthogonal views is embodied by X2CT-GAN [10], which reconstructs an entire CT volume from biplanar X-rays with a generative adversarial network. X2CT-GAN is generative and dense: it hallucinates a full volume, optimizing image-similarity losses. Our problem is discriminative and sparse: we do not reconstruct anatomy, we triangulate a small set of fracture points and must decide

Table 1: Case-disjoint data splits. The diagnostic split is a held-out slice of the development pool used for all out-of-fold selection; the sealed cohort is opened once. Negative denotes fracture-free cases.

Split	Cases	Fract.	Neg.	Purpose
Training pool	325	2,686	20	model training
Diagnostic (held out)	65	492	4	OOB selection
Sealed test	55	601	0	confirmation

correspondence explicitly. This distinction matters for evaluation, since a generative volume can look plausible while a triangulated point is either correct within tolerance or not. Our study can be read as isolating the correspondence bottleneck that a sparse biplanar reconstructor faces, a bottleneck a generative model hides inside its learned prior.

Cross-view correspondence and selective prediction. Matching observations across views is a classical geometry problem: with calibrated cameras, epipolar constraints reduce the search for a point’s partner to a line, and one-to-one assignment then resolves the match [3]. Our orthographic setup gives an even simpler constraint, agreement on the shared superior-inferior axis, but as we show that constraint is far weaker than a full epipolar geometry because many distinct fractures share the same axis coordinate. The second relevant thread is selective prediction, where a model may abstain rather than emit a low-confidence output to trade coverage for reliability [2]. Our assignment-with-abstention is exactly such a mechanism at the level of correspondence commitments, and the controlled false-output budget is the coverage constraint under which we evaluate it.

3 METHODOLOGY

3.1 Data and projection

Each CT volume from RibFrac, together with its fracture annotations and RibSeg-derived rib segmentation, is rendered to AP and lateral orthographic projections at 256×256 . The superior-inferior (SI) axis is shared between the two views, which is the geometric key that later gates candidate pairs.

Table 1 summarizes the case-disjoint splits. Splits are patient-disjoint at the CT level: every volume, and therefore every projection, peak, and crop derived from it, belongs to exactly one split, so no same-scan material crosses the development-test boundary. The detector is trained on the development pool; all staged diagnostics, calibration, and policy or model selection are performed out of fold on the held-out diagnostic split (the L2 lateral head is retrained on the development-internal training slice and never sees the diagnostic or sealed cases); and the addressing model is trained on the same development pool. The sealed cohort is opened once, for a single fixed-policy pass. We note explicitly that the sealed cohort contains no fracture-negative studies, so per-case false-output behavior on truly negative scans is not measured.

3.2 Pipeline

Figure 1 shows the workflow. A per-view U-Net [7] detector emits a fracture heatmap for each view; local peaks are extracted with per-view non-maximum suppression (NMS) and a score floor. Peaks are paired into a candidate graph gated by SI agreement ($|\Delta SI|$ below a tolerance). A one-to-one *assignment with abstention* commits a correspondence only when its matching cost clears a threshold, so low-confidence pairs produce no 3D output. Each committed pair is back-projected and triangulated to a 3D point. A separate addressing model predicts the rib level of each detection. The detector, extraction policy, correspondence configuration, and evaluation are all frozen before the sealed test, and every stage verifies data, checkpoint, and protocol hashes before producing any number (fail-closed provenance).

3.3 Networks

The detector (Figure 2) is a from-scratch U-Net with base channel width 32, deployed as two independent copies (AP and lateral), about 1.95M parameters per view. Each stage is a double convolution (Conv 3×3 , BatchNorm, ReLU, twice); the encoder halves resolution with max-pooling and the decoder restores it with bilinear upsampling and skip concatenation, ending in a 1×1 convolution and sigmoid. Training uses a penalty-reduced focal loss [6]. The addressing network is a compact dual-stream CNN (about 48.8k parameters) whose two 64-dimensional view embeddings concatenate into three linear heads (side, rib level, and a quality score); it is included as a supporting component for the review interface and is not the primary evaluated contribution of this study, so we do not report standalone addressing metrics here. The learned pair-scorer, used only to test whether appearance beats geometry for correspondence, is a shared two-layer tower embedding 40×40 crops with a small multilayer-perceptron head (about 12.6k parameters).

3.4 Operational endpoint

From Stage D1 onward we use a single frozen endpoint. For the accepted 3D points in a case, we back-project each accepted AP-lateral pair to a world-space point, compute its distance to every ground-truth (GT) fracture volume (nearest voxel via a per-fracture KD-tree), and run an independent one-to-one prediction-to-GT matching at 5, 10, and 15 mm. Recall at X is distinct GT matched within X mm divided by all GT; a false 3D point is an accepted prediction unmatched at the tolerance. The primary endpoint and false-output budget are at 10 mm and at most one false 3D point per case. This budget is a pre-specified, stringent system-development criterion, not a clinically validated tolerance; it encodes the design goal that a practical reconstructor should rarely place a spurious 3D point, and the analyses below are reported against it rather than against a clinically established threshold.

3.5 Staged diagnostic design

Rather than tuning end-to-end, we run a diagnostic elimination (Figure 3) that swaps one component at a time from

oracle to model, so each experiment charges the failure to a single stage: geometry (Stage A), localization under oracle correspondence (Stage B), the deployed correspondence (Stage C), the candidate graph (Stage D0), deterministic and learned pair-scoring (Stages D1–D2), lateral calibration (Stages L0–L1), and finally the lateral retraining intervention and a detector-by-correspondence factorial (Stages L2–L3).

3.6 Experimental setup and reproducibility

Both detectors are from-scratch U-Nets (base width 32, Batch-Norm) trained with a penalty-reduced focal loss [6] and Adam. The Stage L2 lateral head is warm-started from the frozen lateral weights and trained for 60 epochs with a positive-branch weight of 3.0 and hard-negative emphasis re-mined every ten epochs on the false peaks that survive the extraction policy; the final epoch is taken without validation-based checkpoint selection. Peaks are extracted with AP non-maximum suppression radius 5 and floor 0.05, and lateral radius 3 and floor 0.10 (the standing L2 policy). The assignment introduces a dummy unmatched node at per-node cost u for every real node, so a real edge of cost c is preferred only when $c < 2u$; equivalently, a correspondence commits only when its geometric-mean confidence clears $1 - 2u$ (0.167 at the operating point). We sweep u and the SI gate and select the case-level configuration out of fold on the operational objective (false 3D points at 10 mm), never on the sealed data. Every stage recomputes and checks the SHA-256 of its inputs (dataset, detector checkpoints, frozen policy files) before emitting any number and aborts on mismatch; the sealed evaluation additionally binds to an external data anchor so the test cohort cannot be silently swapped. This fail-closed provenance makes the single confirmatory pass auditable. Confidence intervals are case-level bootstraps with 1,000 resamples over cases. Full training schedules, fold construction, heatmap-target generation, and the frozen policy files are recorded in the project repository so every reported number can be regenerated.

4 RESULTS

4.1 Geometry is exact; localization is not the dominant barrier

Pure orthographic back-projection scored against GT geometry gives a 0.0 mm transform round-trip and, from correct paired GT centers, median 0.0 mm 3D error with 98.6% rib-exact: triangulation is sound (Stage A). Replacing oracle centers with detector peaks while keeping oracle correspondence gives median 4.0 mm, 88% within 10 mm, rib-exact 93.6%, and rib ± 1 100% (Stage B). Localization error is therefore real but tolerable relative to the correspondence failure that follows, with the lateral view the weaker localizer; it is not the dominant barrier, not a claim that localization is perfect.

Biplanar detection and 3D localization pipeline

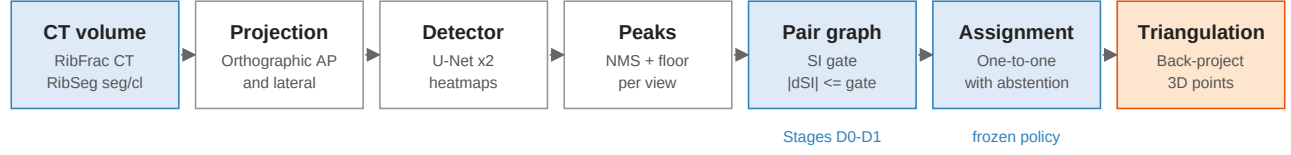
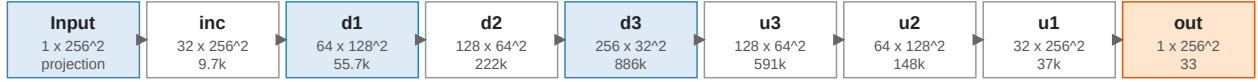


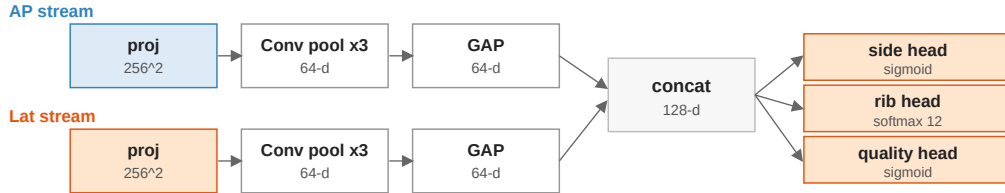
Figure 1: End-to-end pipeline. Two orthographic projections are detected independently, peaks are paired by shared-axis (SI) geometry into a candidate graph, and an assignment with abstention commits only confident pairs for triangulation. The operational bottleneck is that too few true cross-view edges receive enough detector confidence to clear the commit threshold.

(a) Per-view detector U-Net (base channels = 32)

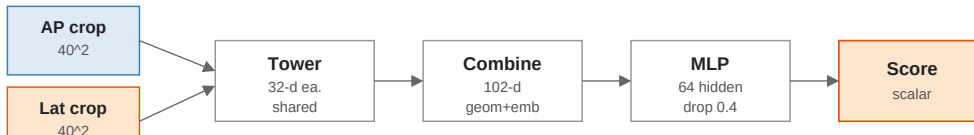


Two independent copies (AP and lateral); ~1.95M params each.

(b) Addressing network (48.8k params)



(c) Learned pair-scorer, D2b (12.6k params)



Pair-scorer used only for attribution; at the operational budget it does not beat detector confidence.

Figure 2: Layer-by-layer architecture of the three trained networks. The biplanar detector is two independent single-channel U-Nets (base channels 32, about 1.95M parameters each); “fusion” is a geometric candidate union on the SI axis, not a learned layer. The addressing network (48.8k parameters) and the learned pair-scorer (12.6k parameters) are the two auxiliary models.

4.2 Correspondence on the frozen detector fails

The deployed SI-only fusion matcher correctly and uniquely pairs only 4 of 205 dual-view fractures, and end-to-end within-10 mm is about 0.6% (Stage C); when a pair is correct

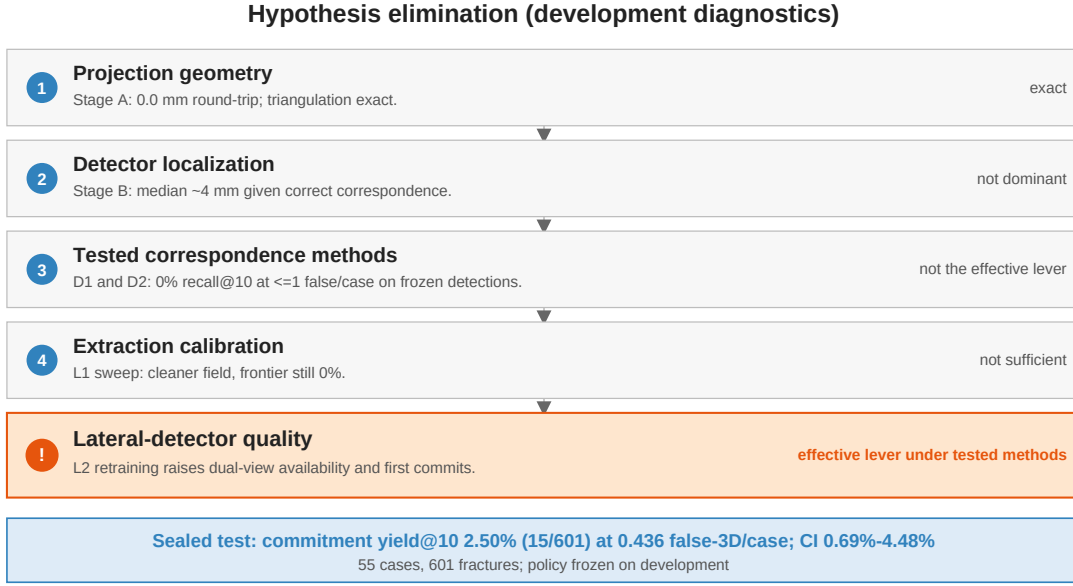


Figure 3: Identifying the effective operational bottleneck through staged elimination. Geometry is exact and localization is not the dominant barrier; the tested deterministic and local-appearance correspondence methods are not the effective lever, and candidate-field density alone is not sufficient; lateral-detector quality is what moves the frontier.

the reconstruction is still good, confirming the failure is in getting correct pairs, not triangulation. The frozen detector emits about 40 AP but about 188 lateral peaks per case, so the SI-gated graph is dominated by non-fracture pairs, and $|\Delta SI|$ cannot separate same-fracture from cross-fracture pairs (identical medians). Dual-view availability is only 51.8%, an upper bound on recall for the current candidate-generation pipeline (Stage D0). One-to-one assignment with abstention reconstructs 0% at the budget, and the learned two-tower appearance scorer is essentially at chance on the hard positive-versus-cross task (AUROC 0.57), so the tested pair-scoring family is operationally exhausted on the frozen detections (Stages D1–D2). Table 2 shows the shape of the failure: there is no gentle operating point. Recall is exactly zero at any tolerable budget and only becomes nonzero once dozens of false points per case are accepted, and even fully unconstrained the min-cost pairing realizes only 8.3% of an already-modest candidate ceiling, meaning it selects the wrong partner for most fractures.

4.3 The flood is a calibration artifact, but recalibration is not enough

The lateral heatmap’s per-case maximum sits near 0.091 (AP near 0.176); its output is compressed just above the 0.05 extraction floor, and 68% of spurious peaks live in the single bin $[0.05, 0.055)$ (Stages L0–L0.1). Re-tuning the lateral extraction policy (NMS radius 5 to 3, floor 0.05 to 0.10) shrinks the candidate field about fourfold and recovers merged-away

Table 2: Deterministic correspondence (Stage D1) operating frontier, out of fold on the development split. The transition from “accept nothing” to “many false points” is abrupt; there is no controlled-budget operating point.

False 3D points / case	Recall @10 mm	Fractures matched
≤ 3	0.0%	0
5	2.4%	12
10	6.3%	31
∞ (accept all)	8.3%	41

availability, yet the operational frontier stays at 0% and is floor-invariant (Stage L1). Table 3 makes the dissociation explicit: across policies the candidate field collapses and dual-view availability rises, while controlled-budget recall does not move off zero. Candidate-field density alone is therefore not the binding lever under the tested policies: what limits the budget is how many true edges the detector is confident enough to commit.

4.4 The detector intervention is the effective lever under the tested methods

Retraining only the lateral head with hard-negative mining on retained false peaks plus positive-branch strengthening lifts

Table 3: Lateral extraction-policy sweep (Stage L1), AP fixed at the deployed policy. Recalibration shrinks the candidate field and lifts availability, but controlled-budget recall stays at zero for every policy. $R@10_{\leq 1}$ is recall at 10 mm at one false 3D point per case; $R@10_{\infty}$ is uncapped.

Policy	Pk/case	Dual-view	$R@10_{\leq 1}$	$R@10_{\infty}$
nms5 / 0.05 (deployed)	187.6	0.518	0.0%	8.3%
nms5 / 0.060	32.0	0.492	0.0%	8.1%
nms5 / 0.065	20.1	0.468	0.0%	7.3%
nms3 / 0.060	44.2	0.589	0.0%	12.2%

Table 4: Detector-by-correspondence factorial (Stage L3): recall at 10 mm out of fold at the one-false-per-case budget, each head at its own calibrated policy. Only the retrained (L2) detector with deterministic assignment moves off zero (uncapped recall in parentheses).

Lateral head	Deterministic (D1)	Learned (D2)
Frozen	0.0% (12.2%)	0.2% (17.3%)
L2 retrained	2.44% (21.5%)	0.0% (20.7%)

the amplitude ceiling (0.091 to 0.155, on par with AP) and dual-view availability (0.52 to 0.76). For the first time the deterministic assignment commits where the frozen head could only abstain, moving the development operational frontier from 0% to 2.44% recall at 10 mm at the one-false-per-case budget (Stage L2). The detector-by-correspondence factorial (Figure 4) attributes the observed operational gain to the detector intervention within the tested 2×2 design: the only cell that moves off zero is the retrained detector with deterministic assignment (Table 4). Learned appearance does not beat detector confidence at the budget; its positive-versus-cross AUROC rises with detector quality (0.57, 0.66, 0.71) but never enough to commit, and on the retrained head the learned scorer is actually worse operationally than the detector’s own geometric-mean confidence (Stage L3). The floor-invariance of the L2 operating point (the same 12 fractures commit across floors 0.06 to 0.12) confirms that the budget is set by how many edges clear the confidence threshold, not by competitor density.

4.5 Sealed-test confirmation

A single pre-specified pass on the untouched 55-case cohort, with the extraction policy and correspondence configuration frozen on development and no test-set reselection, reproduces the result (Figure 5 and Table 5). We read it in pipeline order rather than end-to-end first. *Evidence availability is substantial*: the retrained (L2) detector reaches 61.1% dual-view availability and a 58.4% candidate ceiling at 10 mm, so most fractures are in principle recoverable from the candidate space. *Conditional localization is accurate*: when the

Table 5: Sealed cohort (55 cases, 601 GT fractures, no fracture-negative cases). Frozen detector’s selected controlled-budget policy versus the retrained (L2) detector, under the pre-specified protocol (fixed policy; no test reselection). The commitment yield @10 mm is end-to-end fracture recall under the committed-output policy, with an all-fracture denominator (all 601 GT fractures).

Metric	Frozen detector	L2 detector
Correct 3D commitment yield @10 mm	0.0% (0)	2.50% (15)
Commitment-yield 95% CI	[0.0, 0.0]	[0.69, 4.48]
False 3D points / case	0.0	0.436
Cases with ≥ 1 correct	0 / 55	6 / 55
Matched median distance	–	1.49 mm
Matched rib-exact	–	0.93
Dual-view availability	0.542	0.611
Candidate ceiling @10	0.265	0.584

policy commits a correct pair, median error is 1.49 mm with rib-exact 0.93 and rib ± 1 1.00. *The end-to-end yield is the terminal metric of this funnel*: under the deliberately conservative fixed policy the L2 detector promotes 15 of 601 fractures to correct 3D localizations, a 2.50% correct 3D commitment yield at 10 mm at 0.436 false 3D points per case, with a case-bootstrap 95% interval [0.69%, 4.48%] above zero, whereas the frozen selected policy emits no predictions at all (0 of 601, a degenerate zero interval). That 2.50% is the joint outcome of several selective gates, i.e., detection in both views, inclusion of the true pair in the candidate graph, survival of one-to-one assignment, confidence above the abstention threshold, and localization within tolerance; it is a conservative end-to-end commitment yield, not a measurement of the detector, geometry, or rib addressing in isolation. The result supports nonzero performance under the fixed L2 policy; it is not a paired hypothesis test of the detector difference, though the per-case direction is one-sided (6 cases improve, none worsen). The sealed point estimate was similar to the development out-of-fold estimate (2.50% versus 2.44%), and the detector-side mechanism transferred intact.

4.6 Impact

The primary scientific value of this study is diagnostic, while its application-level value is an assistive-workflow proof of concept rather than a deployable clinical system. It converts a vague “biplanar 3D does not work” into a specific, reproducible claim: geometry and triangulation are exact, localization is rib-accurate, and the operational failure manifests at the cross-view correspondence stage, where the tested methods depend on detector confidence and candidate quality, so the gain came from lateral-detector quality rather than from the tested matching methods. That result redirects effort away from ever-more-elaborate pair-scoring and toward detector quality and global anatomical disambiguation, and

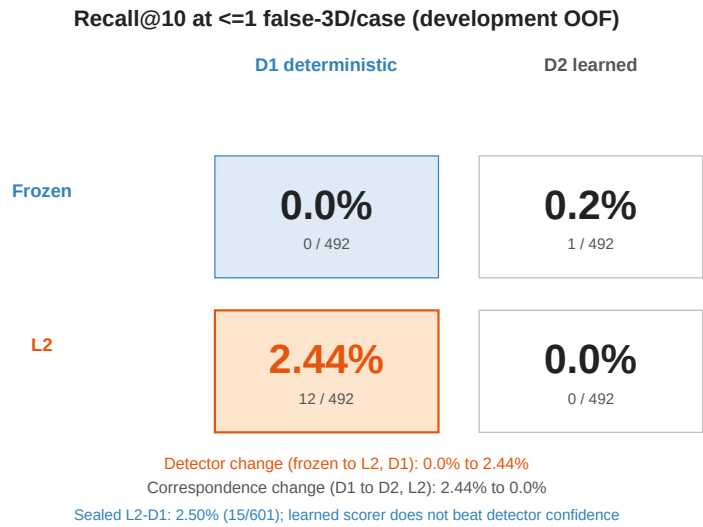


Figure 4: Detector-by-correspondence factorial. Only the retrained (L2) detector paired with deterministic detector-confidence assignment moves off zero; the learned pair-scorer does not beat detector confidence at the operational budget.

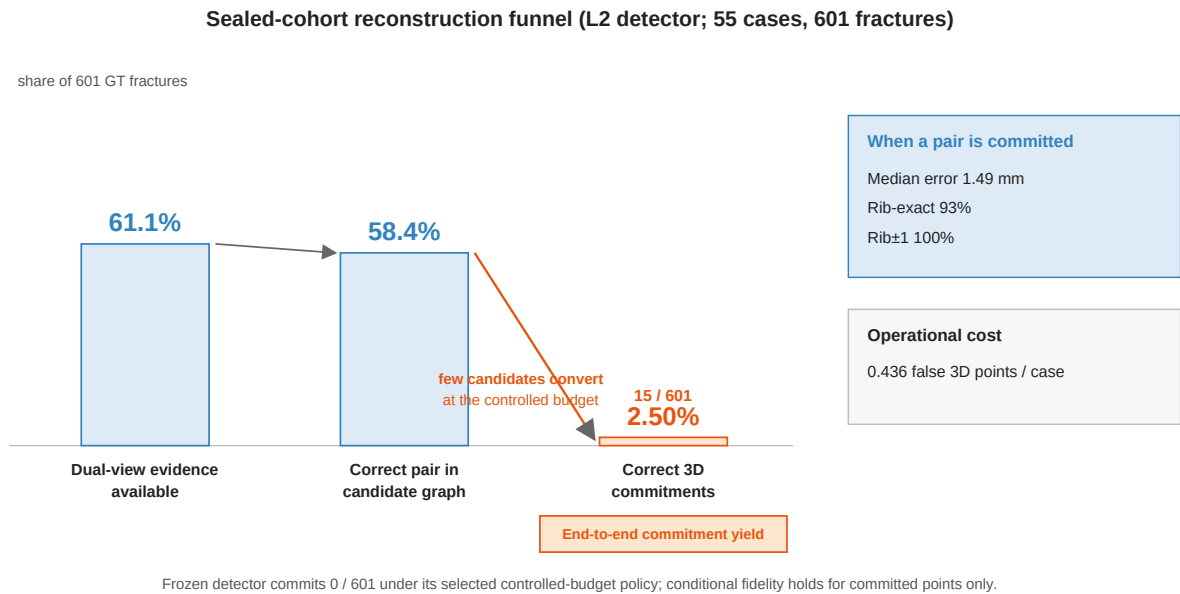


Figure 5: Sealed-cohort reconstruction funnel for the retrained (L2) policy. A large share of fractures reaches a recoverable candidate state (61.1% dual-view available, 58.4% with a correct pair in the candidate graph), but the controlled-budget assignment converts few of these into correct 3D commitments: the end-to-end commitment yield is 2.50% (15 of 601). Committed points are accurate (median 1.49 mm, 93% rib-exact) at 0.436 false 3D points per case; the frozen policy commits 0 of 601.

it does so with a pre-specified sealed confirmation and hash-checked provenance, so the finding is trustworthy despite the modest absolute commitment yield.

At the application level, the system demonstrates a layered assistance model: per-view detections provide highlighting, rib addressing supplies anatomical organization, and only

sufficiently confident AP-lateral pairs are promoted to 3D points, so it offers partial information without fabricating spatial certainty. In selective-prediction terms, coverage is low but conditional localization quality is high (committed sealed points had median error 1.49 mm and 93% rib-exactness), and the lower-level 2D findings remain available when the system abstains. The results therefore support *selective* localization rather than comprehensive reconstruction, and we describe this as potential assistive utility and a workflow proof of concept, not demonstrated clinical benefit, since no clinician reader study has shown improved accuracy or review time.

4.7 Clinical workflow relevance

RibAssist 3D is intended as a secondary-review and anatomical-organization tool for chest-trauma CT, not an autonomous diagnostic system. A potential user is a radiologist, emergency physician, or trauma clinician reviewing suspected thoracic injury: the CT is read conventionally, and the system runs alongside it to highlight suspected fractures, predict side and rib level, and place selective 3D locations, while findings without confident correspondence stay visible rather than being discarded. Plausible applications are secondary fracture review, rib-level documentation, structured injury summaries, and prioritization for specialist review where coverage is limited. These are hypothetical here: with no clinician reader or workflow study, we claim workflow *relevance*, not demonstrated clinical benefit.

4.8 Clinician-review proof of concept

To situate the mechanism in a workflow, we built an interactive review interface (Figure 6) that runs the trained models live. Detection and rib addressing remain the primary output; biplanar 3D localization is an *additive* layer. A detection is never removed because the 3D correspondence abstains: findings are shown in three explicit states, model-localized (a committed, triangulated 3D point), candidate (a cross-view pair that did not clear the commit threshold), and rib-level-only (a 2D detection with addressing but no 3D), so an abstained pair still surfaces as a reviewable finding rather than disappearing. The interface presents the AP and lateral viewers with their markers, the 3D rib anatomy with committed and candidate locations, the detection and addressing confidences, cross-view status, and accept, needs-review, and reject actions, along with provenance and audit information. It is a workflow demonstration, not a usability-validated or clinically validated product, and no clinical decisions were made with it.

5 LIMITATIONS AND THREATS TO VALIDITY

Several constraints bound the strength of the claim. *Simulated projections*: the AP and lateral views are orthographic renderings of CT, not real biplanar radiographs, so scatter, tissue overlap, and calibration error of a real acquisition are not modeled. *Small sealed cohort*: at $n = 55$ the confidence interval is wide and the estimate is directional rather

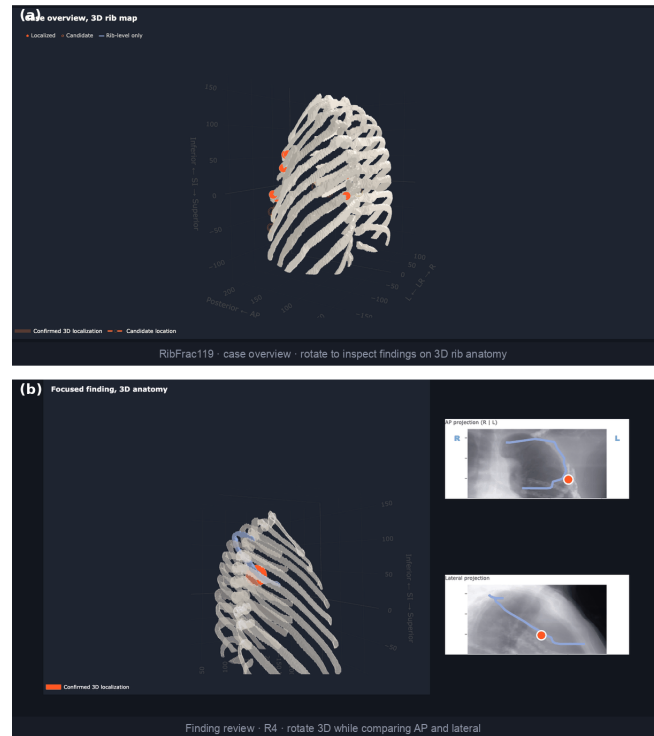


Figure 6: Clinician-review proof of concept (live model outputs). (a) Case overview on 3D rib anatomy, with model-localized, candidate, and rib-level-only findings shown as distinct states. (b) Per-finding review: the AP and lateral viewers with markers beside the 3D anatomy and the committed (localized) 3D point. Detection and addressing remain primary; 3D localization is additive and abstains without dropping a detection.

than precise. *Negative-scan safety unvalidated*: the sealed cohort contains no confirmed fracture-negative studies, so the per-case false-output behavior on truly negative scans is unmeasured, a gap that matters for any triage use. *Development selection*: the retrained head was chosen on the development split, and the sealed pass confirms the direction of the effect but not an unbiased effect size. *Joint mediation*: retraining changed availability, prevalence, confidence, and the candidate ceiling together, so the gain is attributed to detector quality overall, not to availability as an isolated causal mediator. *Metric scope*: recall at 10 mm with independent prediction-to-ground-truth matching rewards localization within tolerance at a controlled false rate, but does not assess fracture typing, displacement, or acute-versus-healed status.

6 ETHICS AND INTENDED USE

This retrospective study uses public, de-identified research datasets (RibFrac and RibSeg v2) under their respective licenses, with no patient interaction and no prospective or

patient-facing decisions. RibAssist 3D is a research proof of concept, not a medical device, and must not be used for diagnosis or patient care. The clinician-review interface demonstrates human-in-the-loop review but has not undergone usability or clinical validation. Because the AP and lateral inputs are simulated CT-derived projections, no claims are made about performance on independently acquired radiographs; no demographic subgroup or fairness analysis was performed because the available dataset metadata and evaluation did not support one, which is itself a limitation for equitable deployment.

7 CONCLUSION AND FUTURE WORK

We presented RibAssist 3D, a staged diagnostic study of biplanar 3D rib-fracture reconstruction from CT-derived projections. By swapping components from oracle to model one at a time, we identified the effective operational bottleneck under the tested detector and correspondence methods: projection geometry is exact and detector localization is rib-accurate, so the operational failure manifests at the cross-view correspondence stage, where weak lateral detections limit confident commitment, and the gain came from lateral-detector quality rather than from the tested deterministic or local-appearance correspondence methods. A lateral retraining intervention produced the first nonzero controlled-budget reconstructions, moving the frontier from 0% to 2.44% in development, and a single pre-specified evaluation on an untouched 55-case cohort confirmed a 2.50% correct 3D commitment yield at 10 mm at 0.436 false 3D points per case, with a bootstrap interval above zero. We additionally implemented an interactive review prototype that preserves all original detections, shows model-localized and uncommitted candidate locations as distinct states, and supports human review without presenting the system as diagnostically validated. The sealed result therefore establishes feasibility for confidence-gated, selective 3D localization and for an assistive workflow, highlighting suspected findings in each view, assigning anatomical rib information, preserving uncertain detections for review, and adding accurate 3D localization only when confidence is sufficient, while identifying the detector and global-correspondence improvements required for broader coverage.

This is a directional proof of concept rather than a complete standalone rib-fracture reconstructor. Its current value lies in fracture highlighting, rib-level organization, and selective high-confidence 3D localization, while its low end-to-end commitment yield (2.5%, with 6 of 55 cases producing any reconstruction) prevents comprehensive automated reconstruction, and it inherits the limitations above. If we were to redo the project, we would budget more effort earlier toward lateral-detector quality rather than toward the correspondence algorithm, since the staged analysis shows that detector quality was the effective lever within the tested system. Importantly, we did not rule out the broader correspondence design space: global anatomical matching, rib identity and

ordering constraints, graph-based or full-view feature correspondence, and jointly learned detection-correspondence models remain open and are the natural place for future gains beyond the local pair-scoring tested here. A cleaner lateral detector that raises confidence on more true edges without the mining-versus-amplitude conflict observed here, additional projection views, real (non-simulated) biplanar inputs, and a 3D confirmation pass over accepted points are further extensions.

CODE AVAILABILITY

Source code, split identifiers, frozen policies, evaluation scripts, and reproducibility instructions are available in the RibAssist 3D repository [9]. Third-party datasets and trained checkpoints are not redistributed because of licensing and artifact-size constraints.

REFERENCES

- [1] Benjamin T. Flagel, Fred A. Luchette, R. Lawrence Reed, Thomas J. Esposito, Kimberly A. Davis, John M. Santaniello, and Richard L. Gamelli. 2005. Half-a-dozen ribs: The breakpoint for mortality. *Surgery* 138, 4 (2005), 717–723. <https://doi.org/10.1016/j.surg.2005.07.022>
- [2] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.
- [3] Richard Hartley and Andrew Zisserman. 2004. *Multiple View Geometry in Computer Vision* (2 ed.). Cambridge University Press, Cambridge, UK.
- [4] Liang Jin, Shixuan Gu, Donglai Wei, Jason Ken Adhinarta, Kaiming Kuang, Yongjie Jessica Zhang, Hanspeter Pfister, Bingbing Ni, Jiancheng Yang, and Ming Li. 2023. RibSeg v2: A Large-scale Benchmark for Rib Labeling and Anatomical Centerline Extraction. *IEEE Transactions on Medical Imaging* 43, 1 (2023), 570–581. <https://doi.org/10.1109/TMI.2023.3313627>
- [5] Liang Jin, Jiancheng Yang, Kaiming Kuang, Bingbing Ni, Yiyi Gao, Yingli Sun, Pan Gao, Weiling Ma, Mingyu Tan, Hui Kang, Jiajun Chen, and Ming Li. 2020. Deep-learning-assisted detection and segmentation of rib fractures from CT scans: Development and validation of FracNet. *EBioMedicine* 62 (2020), 103106. <https://doi.org/10.1016/j.ebiom.2020.103106>
- [6] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2020. Focal Loss for Dense Object Detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 42, 2 (2020), 318–327. <https://doi.org/10.1109/TPAMI.2018.2858826>
- [7] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. Springer, 234–241. https://doi.org/10.1007/978-3-319-24574-4_28
- [8] Mehmet Sirmali, Hasan Türüt, Salih Topçu, Erkmén Gülhan, Ünal Yazıcı, Sadi Kaya, and Irfan Taştepe. 2003. A comprehensive analysis of traumatic rib fractures: morbidity, mortality and management. *European Journal of Cardio-Thoracic Surgery* 24, 1 (2003), 133–138. [https://doi.org/10.1016/S1010-7940\(03\)00256-2](https://doi.org/10.1016/S1010-7940(03)00256-2)
- [9] Kabila Haile Soboka. 2026. RibAssist 3D: Source Code, Split Identifiers, Frozen Policies, and Reproducibility Instructions. Software repository. <https://github.com/kabJhai/RibAssist-3D>
- [10] Xingde Ying, Heng Guo, Kai Ma, Jian Wu, Zhengxin Weng, and Yefeng Zheng. 2019. X2CT-GAN: Reconstructing CT from Biplanar X-Rays with Generative Adversarial Networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10619–10628. <https://doi.org/10.1109/CVPR.2019.01087>
- [11] Douglas W. Ziegler and Naresh N. Agarwal. 1994. The Morbidity and Mortality of Rib Fractures. *The Journal of Trauma* 37, 6 (1994), 975–979. <https://doi.org/10.1097/00005373-199412000-00018>